LearnSync AI: AI Powered Adaptive Learning Application
(LearnQuest Co.)

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CHAPTER I

INTRODUCTION AND ITS BACKGROUND

Since the start of the 21st century, technology has completely reshaped how we do business, communicate, and access services. This change is driven by using data and smart analysis to make systems more efficient, scalable, and user-focused. This push for modernization is forcing every industry to abandon old, rigid ways of operating, creating a perfect environment for innovation in fields that are still stuck in the past.

However, despite rapid technological advancements, many educational institutions still rely on standardized, one-size-fits-all learning tools. For instance, students often use traditional reviewers or quiz apps that give the exact same set of questions to everyone, regardless of their current skill level or specific knowledge gaps. This generic approach often leads to student frustration and wasted study time, as they either review concepts they already know or struggle with material far beyond their comprehension.

AI-driven adaptive learning is fundamentally changing this dynamic. By leveraging artificial intelligence, this method personalizes the learning journey for each student, representing a major shift towards a more efficient and student-centric educational model (Paul, 2025). Studies indicate that adaptive learning systems utilize predictive analytics and real-time data to create highly

personalized learning pathways, effectively addressing individual knowledge gaps (Akgun & Greenhow, 2021; Tretow-Fish & Khalid, 2022).

The core of our AI adaptation is difficulty adjustment and pattern recognition within a low-stakes, independent study framework. The system continually monitors student performance, tracking practice quiz scores, response latencies, and concept-specific mastery levels. If a student consistently demonstrates high proficiency in one topic, the system automatically introduces more challenging questions or new, related concept branches to expand their understanding. Conversely, if a student struggles, the AI offers immediate, targeted reinforcement with fundamental practice questions or constructive semantic feedback before advancing. This personalized approach makes independent study consistently effective and engaging.

The role of technology in modern education extends beyond customizing individual learning paths. These systems are also crucial for sustaining study motivation and facilitating secure peer collaboration. For instance, platforms incorporate gamified elements like experience points (XP), learning streaks, and milestone badges to boost intrinsic motivation and encourage consistent study habits (Balaquiao, 2024; Kassenkhan et al., 2025). Furthermore, in collaborative settings, digital tools enable students to share their customized study materials, review their peers' written explanations, and tackle shared practice sets. Rather than using invasive AI algorithms to monitor private group chats or audit team workloads, structured database-level sharing rules allow students to safely

collaborate, validate each other's work, and build critical-thinking skills within secure, self-regulated peer environments.

LearnQuest is an Educational Technology company that focuses on creating highly engaging and effective digital learning experiences by integrating these advanced technologies. The company operates within the EdTech Industry that positions itself in Adaptive Learning Platforms and Tools to incorporate 21st Century Skills. The company serves dual markets: a B2C (Business to Consumer) model to sell directly to individual students and lifelong learners, and a B2B (Business to Business) model to partner with educational institutions and other companies that are interested in improving educational technology.

LearnSync AI is a Software as a Service (SaaS), cloud-hosted adaptive learning platform designed to deliver a truly personalized educational experience. The system intelligently parses course materials in real-time, dynamically mapping concept hierarchies and generating tailored formative practice questions. To further enhance student engagement, the platform includes the PlayMe module, a gamified portal where students can play content-driven games (such as Crosswords and Matching Cards) procedurally compiled from their chosen study resources, drawing inspiration from interactive platforms such as Quizlet and Gizmos.

Furthermore, the platform features CollabShare, a secure peer-to-peer workspace sharing utility. This module is designed to foster cooperative, self-directed learning without relying on intrusive surveillance algorithms. When a

student chooses to publish their private study space, CollabShare generates a secure sharing token and applies strict PostgreSQL Row-Level Security (RLS) policies on the backend database. This allows authorized peers to access their custom flashcard decks, read their Free Recall and Feynman simplifications, and run their space-generated adaptive tests in a protected, Read-Only format. Peers can also securely clone these shared workspaces to create independent, customizable copies. For educators, a comprehensive analytics dashboard provides valuable insights into class-wide and individual mastery levels across topics, enabling timely academic interventions and supporting both autonomous and guided instruction.

The LearnSync AI system is designed to benefit everyone involved in education. For our company, LearnQuest Co., it positions us as a leader in the fast-growing adaptive learning market. Our unique combination of AI personalization, the gamified PlayMe Module, and the secure, database-enforced CollabShare hub gives us a clear competitive edge. This powerful system supports a strong subscription business model, as it is built on delivering real, measurable results for students. Furthermore, students are the biggest winners. They will see better grades, stay more motivated, and retain what they learn thanks to a truly tailored experience. At the same time, teachers and schools have a powerful analytics dashboard that handles the heavy lifting of tracking progress, allowing them to focus on what they do best: mentoring and providing direct support where it is needed most.

This technology is important because it helps solve the big global challenge of providing personalized and effective education to everyone. Technology makes this possible since it can handle and analyze student data to create learning experiences that fit each person, something that teachers alone cannot do on a large scale. This research is needed to test the effectiveness of a system that combines three key parts: adaptive AI for personalized learning, game-like engagement to sustain participation, and secure, read-only peer workspace synchronization (CollabShare) to support cooperative review. While existing tools often specialize in only one of these three areas, this project intends to demonstrate that integrating all three components into a single platform provides the most effective and equitable path for students to reach their full potential and cultivate the necessary teamwork and problem-solving skills for the modern world.

REVIEW OF RELATED STUDIES

This section reviews existing studies and literature that are foundational to the development of the LearnSync AI system. The review focuses on three core areas: AI-driven adaptive learning, gamification in education, and secure collaborative knowledge sharing, highlighting their individual and combined potential to enhance the learning experience.

Recent advancements in artificial intelligence have changed the way digital learning platforms support students, shifting from uniform lessons to more

personalized learning experiences. Studies show that AI systems can analyze how students answer questions like their accuracy, response time, and progress. Also, adjust activities to match their needs (Akgun & Greenhow, 2021). This kind of personalization helps students focus on the areas where they struggle instead of repeating material they already know (Paul, 2025). These ideas strongly support the goals of LearnSync AI, which aims to guide students through more suitable questions and provide help when they start to fall behind. In this context, modern artificial intelligence serves as a transformative ally by creating effective learning trajectories that seamlessly adapt to a student's specific needs and priorities, revolutionizing the concept of lifelong personalized learning (Bayly-Castaneda et al., 2024).

Building on these advancements, recent studies in early 2025 highlight the shift towards Generative AI (GAI) as a primary engine for personalization. Unlike traditional branching logic, GAI-driven systems can create bespoke instructional content and real-time feedback loops that adapt to a student's immediate cognitive load (Frontiers in Education, 2025). Empirical evidence suggests that such dynamic pathways can cause performance gains of up to 25% compared to static adaptive systems by addressing specific knowledge gaps through targeted, procedurally generated interventions. To make adaptive learning work, many systems rely on learner-modeling approaches that estimate how well a student understands a topic. Methods like Item Response Theory (IRT) and Bayesian Knowledge Tracing (BKT) have been widely used because they give clear and interpretable results that teachers can easily understand (Akgun & Greenhow,

2021). Newer approaches rely on machine-learning models such as decision trees, random forests, and gradient boosting, which can improve predictions about the student's next performance based on their past behavior (Paul, 2025). Combining both traditional and modern techniques can help LearnSync AI achieve a balance between accuracy and explainability, which is important for teachers who need to trust and interpret the system's recommendations. Furthermore, researchers emphasize that independent AI architectures can effectively transform classic, static educational content into dynamic adaptive learning flows, reducing the overall workload on educators while providing unique, customized experiences for each learner (Bobocea et al., 2024).

Several studies have also shown that adaptive learning systems generally help students learn faster and retain information longer compared to standard reviewers (Kassenkhan et al., 2025). These results typically come from comparing pre-tests and post-tests or tracking mastery over time. Researchers also emphasize that it is not enough to measure how well the algorithm predicts performance. What matters most is whether it leads to real learning improvement (Akgun & Greenhow, 2021). This supports LearnSync AI's plan to evaluate both the educational outcomes and the system's technical accuracy when determining whether the tool is effective.

Machine learning plays a critical role in enabling intelligent educational systems, particularly those that aim to deliver personalized and adaptive learning experiences. Studies emphasize that supervised learning models such as Decision Trees, Random Forests, and Gradient Boosting are effective for

predicting learner performance and selecting appropriate difficulty levels based on historical data (Paul, 2025). These models can help LearnSync AI determine whether a student is ready for more challenging content or requires reinforcement activities. Additionally, probabilistic methods like Bayesian Knowledge Tracing (BKT) and parametric models such as Item Response Theory (IRT) have been widely used in adaptive learning systems to estimate a learner's mastery of specific concepts (Akgun & Greenhow, 2021). For sequential analysis such as tracking how students improve over time, research suggests that Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks can capture learning patterns more accurately (Kassenkhan et al., 2025). Incorporating these machine-learning tools into LearnSync AI can enhance its ability to generate precise predictions, adapt difficulty dynamically, and create personalized learning pathways that respond to each student's evolving performance.

Gamification has become a widely adopted strategy in digital learning tools as it helps sustain student motivation and engagement. Points, badges, leaderboards, and progress feedback have been shown to improve both attention and persistence in academic tasks (Tretow-Fish & Khalid, 2022). According to Balaquiao (2024), integrating game-like mechanisms in school-related applications increases learner participation and significantly enhances the learning experience. Recent systematic reviews further corroborate that gamification strongly influences intrinsic motivation by facilitating the assimilation of knowledge and directly improving cognitive and self-learning competencies in

educational environments (Jaramillo-Mediavilla et al., 2024). The PlayMe Module in LearnSync AI leverages these findings by embedding motivational challenges and rewards to maintain student interest while reinforcing mastery.

Collaboration also plays a major role in developing higher-order thinking and teamwork skills. Studies show that students perform better in peer-to-peer settings when they can actively share study materials, review peer insights, and validate collective definitions (Akgun & Greenhow, 2021; Paul, 2025). Rather than relying on intrusive, automated AI trackers that monitor chat logs or calculate workload balances, modern collaborative learning emphasizes student autonomy, secure access controls, and voluntary knowledge replication. This literature directly supports LearnSync AI's CollabShare design, which uses PostgreSQL Row-Level Security (RLS) to establish read-only peer-sharing boundaries and secure space replication, promoting self-regulated study groups, transparency, and peer accountability without tracking student conversations.

Learning analytics and teacher dashboards help translate raw student data into insights that teachers can use. Research emphasizes the importance of dashboards that are clear, easy to use, and that highlight important trends such as struggling learners or common misconceptions (Kassenkhan et al., 2025). For LearnSync AI, this means providing teachers with simple summaries of student mastery and actionable suggestions, rather than overwhelming them with complex data (Akgun & Greenhow, 2021). A well-designed dashboard can support teachers in timely interventions.

Because LearnSync AI will be deployed within a school environment that may have limited data and varying internet access, the system must be designed to work effectively under these constraints. Research shows that smaller datasets work best with simpler and more interpretable models, while deep learning approaches require larger amounts of data to perform well (Paul, 2025). This supports adopting a step-by-step approach. Starting with basic models and gradually improving them as more data becomes available (Akgun & Greenhow, 2021).

Finally, although many studies have evaluated adaptive learning, gamification, and collaboration separately, fewer have explored how these components work together in one integrated system. This gap in the literature gives LearnSync AI an opportunity to contribute new findings by combining personalized learning, meaningful gamification, secure sharing tools, and teacher analytics in a single platform (Tretow-Fish & Khalid, 2022; Kassenkhan et al., 2025). By studying how these elements interact, LearnSync AI can offer insights into what makes digital learning systems truly effective.

Taken together, the reviewed studies provide a coherent blueprint for LearnSync AI: (1) adopt hybrid learner models and independent AI architectures that effectively transform static content into dynamic, personalized trajectories for lifelong learning (Akgun & Greenhow, 2021; Bayly-Castaneda et al., 2024; Bobocea et al., 2024; Paul, 2025), (2) use ensemble and sequence models for robust difficulty selection and temporal prediction (Kassenkhan et al., 2025), (3) design gamification to strongly drive intrinsic motivation and improve cognitive

competencies by rewarding demonstrated mastery (Balaquiao, 2024; Jaramillo-Mediavilla et al., 2024; Tretow-Fish & Khalid, 2022), (4) instrument secure, database-level peer-to-peer workspace sharing (CollabShare) to enforce data integrity and student privacy (Paul, 2025), and (5) pair model metrics with educational outcome measures in evaluation (Akgun & Greenhow, 2021). This synthesis directly shapes LearnSync AI's architecture, selection of machine-learning tools, dashboard priorities, and ethical safeguards.

SIGNIFICANCE OF THE STUDY

The outcomes of this research offer valuable insights and technical frameworks across multiple dimensions of the academic ecosystem. The specific significance of the study is outlined below:

For Educational Institutions and Administrators. This study serves as a practical guide for deploying next-generation learning systems. By demonstrating a secure architecture that separates fixed-difficulty exams from isolated practice spaces, it helps institutions transition away from a patchwork of disconnected applications. This single-platform approach reduces server overhead, lowers maintenance costs, and minimizes the security vulnerabilities by consolidating all application services onto a unified Supabase cloud ecosystem.

For Educators. The platform provides a secure, automated way to evaluate students and upload curriculum files without increasing manual

workloads or compromising grading integrity. Through an isomorphic question-generation layout, instructors can create high-stakes exams where the mathematical core and difficulty remain constant across all students, but the contextual scenario and phrasing are randomized. This neutralizes basic answer-sharing and database lookup cheats. Additionally, the dashboard displays aggregated, real-time class competency trends, enabling educators to quickly identify at-risk learners and customize physical lecture materials.

For Students. As the primary beneficiaries, students gain access to a balanced digital learning environment that protects fair grading while supporting personal growth. Within their private Study Spaces, students can access four formative learning modes: manually or AI-generated Flashcards, a Free Recall evaluator with semantic AI feedback, Feynman explanation archives, and a BKT-driven Adaptive Test Mode. Students can self-pace their review sessions and tackle games in the PlayMe module that are compiled *exclusively* from their chosen study files, earning XP, keeping up daily study streaks, and unlocking digital badges. Additionally, students can use CollabShare to securely clone and study peer resources under database-enforced read-only protections, creating a collaborative, self-regulated review hub.

For the Researchers. This study provides a hands-on opportunity to design, deploy, and evaluate a full-stack SaaS application using advanced machine learning and natural language processing. It allows the researchers to bridge the gap between academic theory and real-world software engineering, producing concrete evidence on how to balance automated personalization,

gamification, and secure peer-to-peer workspace synchronization within a single, secure educational platform.

STATEMENT OF THE PROBLEM

Traditional educational methodologies often rely on a rigid, standardized pace that fails to accommodate individual learning curves. Furthermore, many digital learning platforms struggle with academic integrity during online examinations and lack tools that give educators clear, real-time insights into student engagement and group collaboration dynamics. To address these challenges, LearnQuest Co. aims to design and validate LearnSync AI, a cloud-native Software-as-a-Service (SaaS) application. The core engineering problem is to build a unified, secure system that strictly separates formal classroom testing from independent study ensuring that official exams remain fair and standardized while the student practice environment remains highly personalized and adaptive.

To evaluate the platform's technical accuracy, educational effectiveness, and user acceptance, this study will address the following specific research questions:

1. How can natural language processing (NLP) and machine learning be utilized within LearnSync AI to generate fixed-difficulty, isomorphic exam

variants for educators, while securely capturing post-exam student performance telemetry to update adaptive routing profiles inside the autonomous student practice space?

2. How can the PlayMe module's gamification features (experience points, milestone badges, and daily streaks) be structured to increase student motivation and encourage consistent, self-paced study habits?

3. How can the CollabShare synchronization engine facilitate secure, read-only peer-to-peer workspace replication and asset-sharing among students using PostgreSQL Row-Level Security (RLS) to preserve data integrity and prevent unauthorized modifications?

4. How effective is the autonomous adaptive practice space (incorporating Flashcards, Free Recall AI similarity analysis, Feynman explanations, and BKT adaptive testing) in improving student proficiency and concept mastery based on pre-test and post-test performance metrics?

5. What is the level of quality and acceptability of the LearnSync AI application among students and educators when evaluated against the ISO/IEC 25010 international standard in terms of Functional Suitability, Usability, Performance Efficiency, and Security?

SCOPE AND LIMITATIONS

This research focuses on the design, development, and pilot evaluation of LearnSync AI, a cloud-hosted adaptive learning application. The study aims to produce a functional Minimum Viable Product (MVP) to measure the effectiveness of AI-driven personalization, gamification, and collaboration tools in an actual educational setting.

Scope of the Study:

- **Research Focus -** This study focuses on the engineering, development, and validation of the LearnSync AI SaaS platform. It covers cloud microservices, database optimization, and the core algorithms powering the system. To protect grading integrity, the system is designed to separate fixed-difficulty instructor exams from the adaptive student practice environment. System testing will evaluate software quality using ISO/IEC 25010 standards, specifically focusing on Functional Suitability, Usability, Performance Efficiency, and Security.

- **Target Platform -** The system is built as a responsive, cross-platform web application. It is optimized for standard desktop and modern mobile web browsers, ensuring easy access without

requiring any local software installations.

- **User Population & Setting -** The pilot deployment and evaluation will be conducted entirely at the Pamantasan ng Lungsod ng Pasig (PLP) campus. The target participants include a minimum cohort of 100 undergraduate IT and Computer Science students, along with 3 to 5 faculty members who will evaluate the instructor-facing features.

Modules Included in the Design:

- **Adaptive Learning Engine -** This module manages how course materials are processed and delivered across two separate tracks:
  - **Instructor Tracking -** The AI parses faculty-uploaded materials to generate parallel, isomorphic exam questions at a fixed difficulty. While the exam remains unyielding during execution, the system captures student accuracy and response latency upon submission.
  - **Student Tracking -** Restricted entirely to the Student Study Space. This track ingests historical telemetry from previous instructor exams alongside real-time practice data. It

leverages Bayesian Knowledge Tracing (BKT) across four cognitive study modes (Flashcards, Free Recall AI semantic evaluation, Feynman simplifications, and Adaptive Tests) to dynamically adapt quiz difficulty and target identified student weaknesses.

- **PlayMe Module -** This module handles game-like elements to keep students motivated during independent study. It operates on automated rules triggered by user activity. When a student chooses a crossword or match game, the engine dynamically extracts vocabulary definitions exclusively from their chosen learning resources. Features are limited to calculating and displaying experience points (XP) based on accuracy and speed, validating milestone badges, and tracking consecutive daily login streaks.

- **CollabShare Module:** This module facilitates secure, peer-to-peer workspace sharing and database-level synchronization. When a student publishes a study space, the engine generates a secure sync key and applies PostgreSQL Row-Level Security (RLS) on Supabase. This grants viewing peers strict read-only access to custom flashcards, Free Recall/Feynman explanations, and adaptive tests. Peers can clone these rows into their unique user accounts, creating independent copies while leaving the original

space unaltered.

Limitations of the Study:

- **Short-Term Pilot Duration -** Data collection is limited to a pilot testing period of 2 to 4 weeks. Because of this brief timeframe, long-term knowledge retention, multi-year system maintenance, and permanent changes in student behavior remain outside the scope of this study.

- **Limited Content Library -** LearnSync AI is a supplemental study tool, not a replacement for a university-wide Learning Management System (LMS). The platform does not have a pre-built curriculum database. Instead, its content generation depends entirely on the specific reference documents uploaded week-by-week by the 3 to 5 pilot instructors.

- **Algorithmic Cold-Start -** In the autonomous student practice space, the machine learning models face a "cold-start" whenever a new module is uploaded. Because there is no historical data yet, the AI must first generate baseline questions. The system can only start analyzing and adapting to a student's actual understanding

level once they begin answering these initial questions.

- **Technical Dependencies -** The system requires a stable internet connection at the testing venue to sync live telemetry, stream active game sessions, and update adaptive practice paths over WebSockets. Because the storage is consolidated on Supabase PostgreSQL, the application utilizes asynchronous database pipeline triggers to prevent data loss during minor network drops; however, a fully offline mode is excluded from this deployment phase.

- **Sample Size & Generalization -** The pilot study conclusions are based on a sample size of 100 undergraduate IT/CS students and 3 to 5 faculty members from a single institution (PLP). Consequently, findings regarding student behavior, adaptive routing, and engagement may not be fully generalizable to non-technical courses or other universities without further research.

- **Exclusion of Commercial Features -** Although LearnQuest Co. plans to market the platform to B2B and B2C sectors, this study focuses strictly on educational efficacy and software quality. Commercial components (such as credit card payment gateways,

automated subscription billing, and enterprise server auto-scaling) are entirely excluded from this version.

CHAPTER II

INTRODUCTION

This chapter outlines the methodology for designing, developing, and evaluating LearnSync AI as a shared, multi-tenant SaaS application under LearnQuest Co. It presents the engineering frameworks, architectural assumptions, and computational techniques that separate fixed-difficulty isomorphic exams from an autonomous adaptive student practice environment. The chapter details the secure three-tier architecture, Supabase Row-Level Security (RLS) isolation, high-dimensional vector embeddings, and machine learning algorithms needed to unify the platform's core components.

A primary focus is explaining how the system maintains grading integrity while enabling personalized, self-paced learning. Formal instructor assessments remain structurally fixed to ensure classroom fairness, but the platform securely captures post-exam performance telemetry. This automated feedback loop updates each student's latent mastery profile via Bayesian Knowledge Tracing (BKT), allowing future practice sessions to be customized without altering official grades. Beyond individual adaptation, this chapter details the PlayMe procedural game engine for study motivation and the CollabShare peer-to-peer synchronization engine for read-only workspace cloning.

Furthermore, this chapter establishes the operational framework for institutional validation and empirical evaluation at Pamantasan ng Lungsod ng Pasig (PLP). The validation protocol combines black-box testing of security

boundaries with an end-user evaluation that measures software quality against ISO/IEC 25010. Ultimately, this methodology provides a reliable, auditable, and replicable blueprint showing how LearnSync AI safely transforms raw interaction telemetry into actionable, real-time insights for both students and educators.

RESEARCH DESIGN

This section defines and justifies the research framework for evaluating the platform. The study uses a quantitative design combining descriptive-evaluative and quasi-experimental methodologies to assess technical performance, algorithmic accuracy, and educational impact of LearnSync AI in a live academic environment. Rather than subjective manual tracking, this design captures automated backend logs and empirical test data for objective analysis. As noted in recent analyses of AI-driven educational software, quantitative methodologies provide the structural controls needed to measure real-time adaptive adjustments and evaluate statistical correlations across dense telemetry data streams (Wang et al., 2023).

This methodology targets two main data collection areas. The quasi-experimental component measures pre-test and post-test score differences to evaluate how the autonomous practice space improves concept mastery. Simultaneously, the descriptive-evaluative component monitors automated telemetry (correctness ratios, response latencies, game progression, and workspace cloning frequencies). This dual quantitative approach eliminates

human bias when measuring user progress. By transforming raw interactions into numeric data, the study ensures that core functions, performance limits, and Row-Level Security controls can be statistically verified against standardized metrics (ISO/IEC, 2023).

ARCHITECTURAL FRAMEWORK

LearnSync AI uses a secure three-tier model optimized for role-based delivery, personal study spaces, and adaptive algorithms. The system decouples official summative assessments from private formative practice spaces. Official exams remain controlled and standardized, while behavioral telemetry from both classroom tests and independent review continuously optimizes the personal adaptive engines.

1. **Presentation Layer -** The primary user entry point, operating under strict Role-Based Access Control (RBAC) to enforce security and permission partitioning. It is divided into two portals:

    a. **Student Portal:** An interactive environment for self-paced study and engagement. It includes three core components:

        i. *Study Space Hub* - Lets students create private, customizable spaces linked to classes to fetch instructor

resources.

ii. *PlayMe Module Tab* - A gamification portal with content-driven games like crosswords or match games.

iii. *CollabShare Hub* - A peer-to-peer workspace sharing, allowing peers to safely access flashcards, Free Recall logs, Feynman explanations, and space-generated practice tests.

b. **Instructor Portal:** Designed exclusively for class management and performance tracking. Instructors can create classes, manage enrollments, upload unstructured resources (PDFs, PPTs, documents) to classroom feeds, generate official quizzes using anti-chOur unique combination of AI personalization, the gamified PlayMe Module, and the collaborative CollabAI feature gives us a clear competitive edge. eating isomorphic question layouts, and access the Mastery Analytics Dashboard to track individual and class-wide mastery levels.

2. **Application Layer -** The application layer processes all core logic including semantic parsing, AI evaluations, procedural game generation, and database interactions. All client payloads go through a central API Gateway for sanitization, JWT authorization, and RBAC validation before

reaching specialized backend services.

a. **Parsing Engine:** Triggered by instructor uploads or student requests for AI study tools. Using NLP, it maps the conceptual hierarchy of files across two pathways:

    i. *Summative Isomorphic Pathway (instructor-facing)* - generates parallel exam variants with identical difficulty but altered scenarios, names, and values to prevent cheating.

    ii. *Formative Generative Pathway (student-facing)* - parses study resources to extract key terms, generate definitions, and assemble adaptive practice questions.

b. **Cognitive Review Core:** Operates exclusively within the Student Study Space with four interactive modes:

    i. *Flashcard Service* - Supports manual deck creation or AI-driven extraction of key concepts from fetched materials.

    ii. *Free Recall AI Analyzer* - Accepts raw student text, uses semantic embeddings to calculate similarity with reference materials, and returns a coverage score with localized

feedback.

iii. *Feynman Technique Orchestrator* - Extracts foundational concepts, prompts simple explanations, and stores responses as indexable text for review and peer validation.

iv. *Adaptive Test Mode (BKT Engine)* - Generates personalized practice tests using a Bayesian Knowledge Tracing (BKT) microservice that dynamically adjusts question difficulty based on real-time mastery.

c. **PlayMe Engine:** A dual-pipeline gamification engine that operates independently to preserve the integrity of student tracking without altering core BKT mastery metrics. When a student selects a game type and a learning resource, the engine dynamically extracts terms, definitions, and glossaries exclusively from that material, mapping them into a valid crossword grid or randomized associative pairs. Upon game completion, it evaluates performance telemetry to calculate XP, update study streaks, check badge achievements, and write these engagement metrics back to the database.

d. **CollabShare Sync Engine:** A secure peer-to-peer utility that manages access tokens and permissions when a student shares a study space. It enforces access rights using PostgreSQL Row-Level Security (RLS) policies on Supabase, allowing authorized peers to read or clone study assets (flashcards, recall logs, practice tests) into their own portals under a strict, system-enforced read-only condition, preventing unauthorized modification of the original workspace.

3. **Data Tier -** Securely partitions, caches, and isolates multi-tenant data across institutions. It manages three storage vectors:

    a. **Relational Database:** Standard PostgreSQL tables with Row-Level Security (RLS) to manage user profiles, class enrollments, study spaces, shared space metadata, and immutable exam score sheets.

    b. **Semantic Vector Storage:** Native PostgreSQL handling via the pgvector extension, storing high-dimensional semantic embeddings of instructional resources.

c. **Persistent File Storage:** Supabase Storage to securely serve and manage uploaded class resources.

d. **Real-time & Cache:** Supabase Realtime Channels using WebSockets to stream changes directly from the PostgreSQL Write-Ahead Log, keeping games and shared workspaces instantly synchronized.

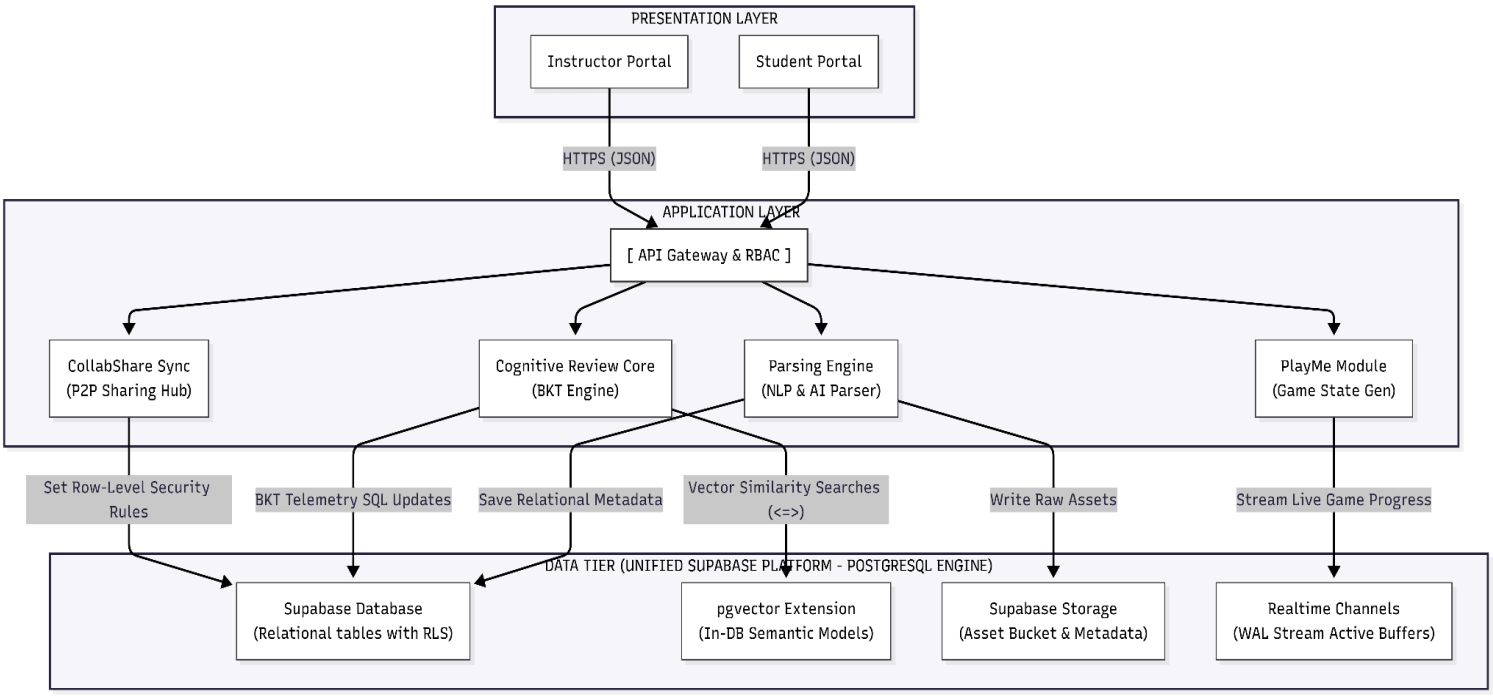


Figure 1. *Architectural Framework*

ALGORITHM

LearnSync AI's core processing layer uses a hybrid framework that manages semantic parsing, cognitive tracking, procedural game generation, and peer-to-peer workspace replication. These duties are organized across four specialized, decoupled engines within the application layer.

- **Document Parsing and Isomorphic Exam Generation -** The content ingestion pipeline starts when an instructor uploads learning resources (PDFs, slides, handouts) to a classroom feed. The system's Parsing Engine uses NLP to extract text, identify core terminology, and build a hierarchical concept map of the lesson's glossary. This parsed metadata then feeds into two distinct application tracks governed by strict database safeguards.

  - **Summative Isomorphic Exam Engine:** To prevent cheating during high-stakes testing, this engine generates parallel, functionally identical exam variants. It maintains the same cognitive load and difficulty for every student while dynamically altering scenarios, names, and values, forcing analytical processing over rote answer-key memorization.

  - **Practice Generation Loop:** Automatically creates fresh baseline question pools for the student's independent practice environment.

This ensures practice materials are immediately available when new reference material is introduced, resolving the algorithmic cold-start problem.

- ➢ **Immutability Rules:** To protect grading integrity, all summative exams and their variants are stored as uneditable, locked records in PostgreSQL. Post-submission modifications are prohibited by the application layer to prevent data tampering.

- ❖ **Hybrid Adaptive Learning Engine (Student Space Modes) -** After creating a private study space linked to an enrolled class, students can access four self-paced formative modes that measure and improve concept retention without affecting official grades:

  - ➢ **Mode A: Flashcard Synthesis Engine:** This service lets students manually build flashcard decks or use a generative script that scans the linked resource's glossary to auto-construct term-definition flashcards, cached locally for rapid active recall.

  - ➢ **Mode B: Free Recall AI Semantic Evaluator:** Prompts students to write a free-form summary. The input is converted into high-dimensional vectors, then compared against the reference document using pgvector cosine similarity. The system returns a

conceptual coverage percentage, highlights recalled topics, and provides localized feedback.

- ➢ **Mode C: Feynman Technique Explanation Core:** Students explain complex terms in simple language. The engine retrieves concept nodes, archives the simplifications, and supports deep cognitive organization. Logs can be saved for personal or peer review.

- ➢ **Mode D: BKT-Driven Adaptive Testing Mode:** A personalized practice loop. The engine monitors correctness and response latency, sends telemetry to a Bayesian Knowledge Tracing (BKT) service, which updates the student's latent mastery level. An adaptive router then selects the next question to keep challenge within the student's learning threshold.

- ❖ **CollabShare Space Engine -** Operates as a secure, non-AI peer-to-peer workspace synchronization utility backed by database-level access rules:

  - ➢ **Secure Token Generation:** When a student publishes a study space, the engine generates a unique synchronization key mapping shared assets to a public directory on Supabase.

- **Row-Level Security (RLS) Permission Gate:** PostgreSQL RLS policies grant viewing peers strict read-only access. They can study flashcards, read Free Recall and Feynman explanations, and run adaptive tests, but cannot modify or delete the owner's original documents.

- **P2P Workspace Replication:** If a peer chooses to clone a shared workspace, the engine duplicates relational rows (e.g., flashcard decks, practice tests) and writes them to the peer's profile under their user ID, creating an independent, customizable copy.

- **PlayMe Game Engine -** The primary gamified incentive engine of LearnSync AI. It operates independently of BKT cognitive metrics to preserve formal study tracking integrity. Structured into two components:

  - **Procedural Game Board Generation:** When a student selects a game (e.g., Crossword, Matching Cards) and links a study space, the engine retrieves the vocabulary glossary from that resource. It filters terms by length, places the longest word as the grid's central anchor, and uses a backtracking layout algorithm to find valid overlapping character intersections. Each board is dynamically tailored to the student's current material.

- **Gamification Reward Calculations:** Upon game completion, the engine evaluates performance telemetry (time spent, correctness, word difficulty) to distribute engagement incentives:

    - **Experience Points (XP) Scaling:** Awards XP proportionally to concept complexity and solving accuracy, using a non-linear factor that rewards fast solvers without penalizing slower readers.

    - **Deterministic Study Streak Tracking:** Captures session timestamps to encourage daily habits, incrementing the streak with continuous activity or resetting it after 48 hours of inactivity to reinforce consistent study.

    - **Milestone Badge Validation:** Scans historical profile metrics (lifetime XP, completed games, BKT mastery) to automatically award digital badges upon reaching preset milestones.

TECHNOLOGY CONFIGURATION

The production architecture separates responsibilities across decoupled systems for high availability, fault tolerance, and maintainability.

- **Frontend Client Interface:** Built with React for a responsive, cross-browser application. It handles local state synchronization, renders PlayMe gamification components, manages card-flipping transitions, and uses the Supabase JavaScript Client to subscribe to real-time sync states.

- **Backend Server Application:** Managed via Node.js/Express or Laravel. Acts as a lightweight, non-blocking API gateway for multi-tenant routing, input validation, and RBAC operations via the Supabase Node Admin SDK.

- **Machine Learning Microservices:** Built with Python, running in a scalable API container. Dedicated to cognitive metric evaluation, NLP document parsing, and BKT updates. Communicates with Supabase Database using SQL to compute cosine similarity via native pgvector.

- **Data Storage & Caching Layer:** Consolidated entirely on Supabase (PostgreSQL). Features schema partitioning, automated database triggers, RLS data security, integrated object storage for raw PDFs, and Realtime WebSocket streaming channels (eliminating external Redis servers).

SE PARADIGM

This study uses Agile Scrum to manage development, testing, and deployment of LearnSync AI. Because the platform combines NLP, adaptive practice, gamification, and peer-to-peer sharing, a rigid linear model is impractical. Empirical studies show sequential models cannot handle the unpredictable design and training of machine learning microservices, whereas Scrum's iterative sprints provide flexibility for continuous tuning and database optimization (Jali et al., 2026; Daraojimba et al., 2024). This adaptive approach is critical for educational platforms where complex frontend telemetry must connect seamlessly with cognitive algorithms within a short development window (Akgun & Greenhow, 2021; Jing et al., 2023).



Figure 2. *Agile SDLC*

The execution of the Agile Scrum framework across the development lifecycle is structured into the following sequential phases:

1. **Phase 1: Product Backlog -** The team defines the master list of requirements from the user's perspective:

    a. **Instructor Core:** Instructors need to upload PDFs or documents so the system automatically generates parallel, fixed-difficulty exam variants to prevent cheating.

    b. **Student Core:** Students need an independent study space with four review modes (Flashcards, Free Recall, Feynman explanations, and Adaptive Practice Tests) where BKT adjusts question difficulty.

    c. **PlayMe Module:** Students need practice scores and game completion to instantly update experience points (XP), daily streaks, and digital badges to sustain motivation.

    d. **CollabShare Module:** Students need the ability to securely share their study spaces with peers to synchronize flashcards, written recall explanations, and practice tests in a

read-only format.

2. **Phase 2: Sprint 1 (Core Infrastructure & Supabase Data Tier) -** This sprint builds the database backend and security boundaries on Supabase:

    a. **Database Setup:** Configure tables on Supabase. Enable and enforce Row-Level Security (RLS) policies in PostgreSQL, keeping tenant data completely isolated between different school client profiles.

    b. **NLP Deployment:** Deploys the Python-based NLP engine to extract text from uploaded documents, generate raw vectors, and store them in Supabase using the pgvector extension.

    c. **Data Protection:** Implements system rules that save all generated quizzes and isomorphic exams as uneditable, immutable database records to protect grading integrity.

3. **Phase 3: Sprint 2 (Engagement & Sharing Engines) -** This sprint develops the behavior-tracking logic and automated reward systems:

a. **PlayMe Engine:** Codes the procedural crossword grid generator and configures Supabase Realtime Channels to stream active game states to the React client.

b. **CollabShare Engine:** Configures the database-level RLS policies on Supabase, establishing read-only access layers and triggers for secure space cloning.

c. **Recall Semantic Evaluator:** Integrates SQL queries using pgvector cosine similarity operators to evaluate unstructured student inputs in the Free Recall review mode natively inside PostgreSQL.

4. **Phase 4: Sprint 3 (Portal Integration & Security) -** This sprint unifies individual components into a single responsive web platform:

a. **API Gateway:** Routes all data traffic through a central gateway that handles input validation, JWT token extraction, and Supabase client authorization checks.

b. **User Interfaces:** Develops the responsive Student Portal and Instructor Portal using React, subscribing to real-time

changes via Supabase Client WebSockets.

c. **Performance Optimization:** Implements database indexing on vector and relational tables, and sets up transaction buffers on Supabase to maintain swift response times during concurrent submissions.

5. **Phase 5: Testing, Pilot Deployment, and Evaluation -** This final phase validates software readiness in a live academic environment:

   a. **System Testing:** Conducts black-box testing to verify Supabase storage access rules and that RLS policies successfully prevent student roles from accessing unauthorized records.

   b. **Pilot Launch:** Deploys the finalized web application for a 2-to-4-week pilot study at Pamantasan ng Lungsod ng Pasig (PLP).

   c. **Quality Evaluation:** Tests the platform with 100 undergraduate IT/CS students and 3-5 faculty members. Users complete an ISO/IEC 25010 survey measuring functional suitability, usability, performance efficiency, and

security.

DATA FLOW DIAGRAM

This section defines the operational boundaries and data transactions of LearnSync AI using standard Data Flow Diagrams (DFDs). These diagrams trace multi-tenant data from initial ingestion through isolated processing and storage to secure, role-based visualization dashboards.

1. **Level 0: Context Diagram -** Establishes the system's overall scope by representing LearnSync AI as a single, centralized cloud process (Process 0.0). This macro-level view maps high-level bidirectional flows between the system boundary and two external entities:

    a. **The Instructor:** Acts as administrative content manager. Inputs unstructured course materials (PDFs, documents) and classroom metadata (enrollments, exam parameters). Receives role-validated class analytics, competency trends, at-risk alerts, and exam performance histories.

    b. **The Student:** Serves as the primary operational user. Feeds interactive data streams (practice answers, free-form recall summaries, Feynman simplifications, game inputs, workspace

sharing/cloning commands). Receives dynamically adjusted formative quizzes, localized semantic study paths, cloned peer workspaces, and real-time PlayMe updates (XP, badges, login streaks).

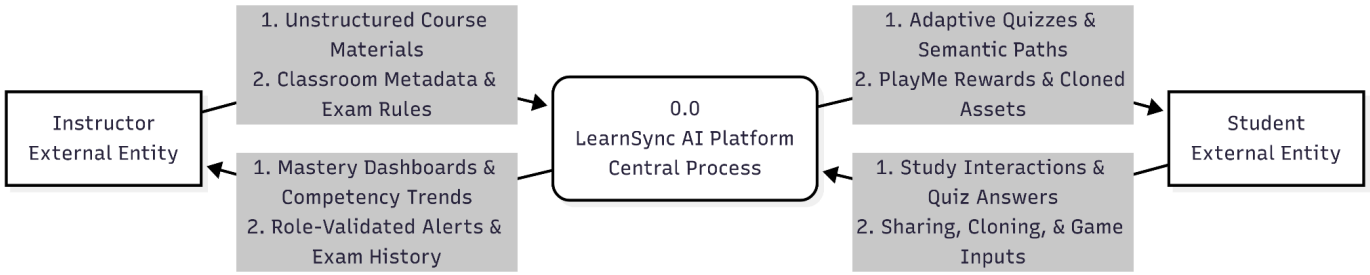


Figure 3. *Level 0 Context Diagram*

2. **Level 1: Main Process -** This diagram decomposes the central LearnSync AI system into six core functional processes, mapping the specific data routing and transformation stages across the unified Supabase storage infrastructure:

    a. **1.0 User Auth & RBAC Guard:** Primary security gateway. Ingests login credentials, validates tenant identities against D1 (Relational Database), and issues JWTs to enforce row-level data visibility.

    b. **2.0 Document Parsing:** Handles instructor resource uploads. Applies NLP to parse PDFs or slides, extracts vocabulary hierarchies, saves raw files to D3 (Persistent File Storage), and

writes semantic definitions to D2 (Semantic Vector Database).

c. **3.0 Immutable Quiz Generation & Adaptation:** Core evaluation layer for the four student learning modes. Queries D2, manages Free Recall AI via cosine similarity, handles Feynman logging, runs BKT adaptive quizzes, commits uneditable performance strings to D1, and triggers the gamification subsystem.

d. **4.0 PlayMe Engine:** Triggered dynamically during formative study. Retrieves vocabulary from D1/D2 for procedural game grid calculations, evaluates session completions, updates parameters via D4 (Real-time & Caching), and logs XP, 48-hour streaks, and badges into the student's relational profile.

e. **5.0 CollabShare Synchronization Engine:** Non-AI peer-to-peer workspace replication utility. When a student creates a public link, the engine enforces PostgreSQL RLS policies in D1, allowing authorized peers to read or clone study sets, flashcards, and practice strings under strict read-only conditions.

f. **6.0 Secure Analytics Formatter:** Reporting service for instructor dashboards. Extracts scoring histories from D1 and semantic progress from D2, restructuring data into decrypted class analytics,

competency trends, and automated at-risk alerts.

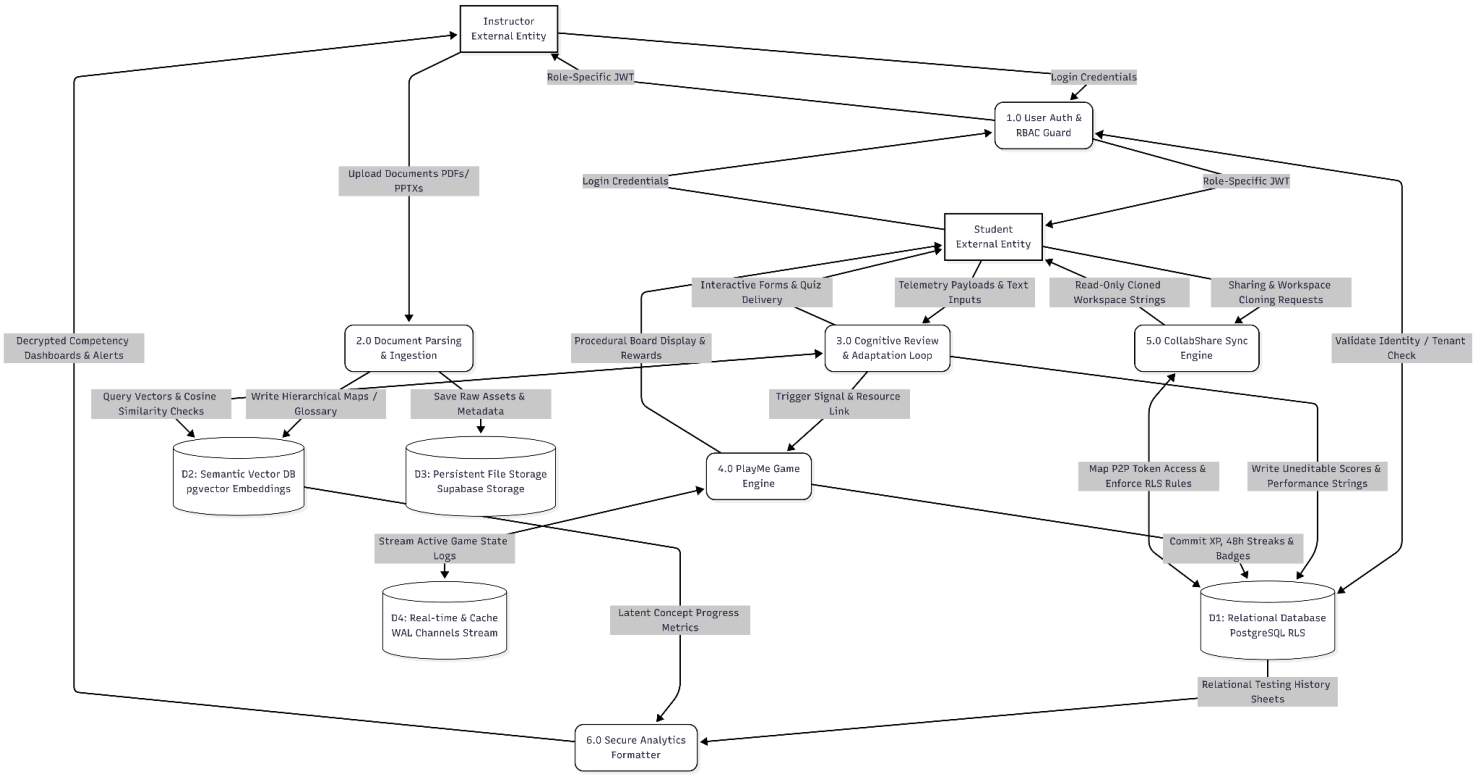


Figure 4. *Level 1 Main Process*

3. **Level 2: AI Detailed Process -** diagram decomposes Process 3.0 (Cognitive Review & Adaptation Loop), isolating the core feedback loop for independent practice testing. It tracks how real-time student inputs are evaluated and transformed into adaptive study paths through four sequential sub-processes aligned with Supabase architecture:

a. **3.1 Evaluate Response Telemetry:** Initiates when a student submits a practice answer. Captures the raw payload, extracts item correctness and time-on-task latency, then routes parsed fields to the tracking engine.

b. **3.2 Update BKT Mastery:** Mathematical core of the adaptive loop. Ingests performance metrics, executes Bayesian Knowledge Tracing (BKT) to recalculate latent knowledge state, and updates mastery vectors within D1 (Relational Database).

c. **3.3 Next Sequence:** Uses calibrated mastery scores to determine the next pedagogical step. Queries D2 (Semantic Vector DB with pgvector embeddings) for conceptual dependencies and prerequisite hierarchies, then routes to the optimal concept node within the student's cognitive threshold.

d. **3.4 Serve Adaptive Question & Lock Record:** Compiles a unique, randomized multiple-choice item for the target concept. Immediately commits the instance as an uneditable, locked record to D1. Concurrently, serves the question payload back to the Student Portal via WebSockets to refresh the interface and sustain the learning cycle.

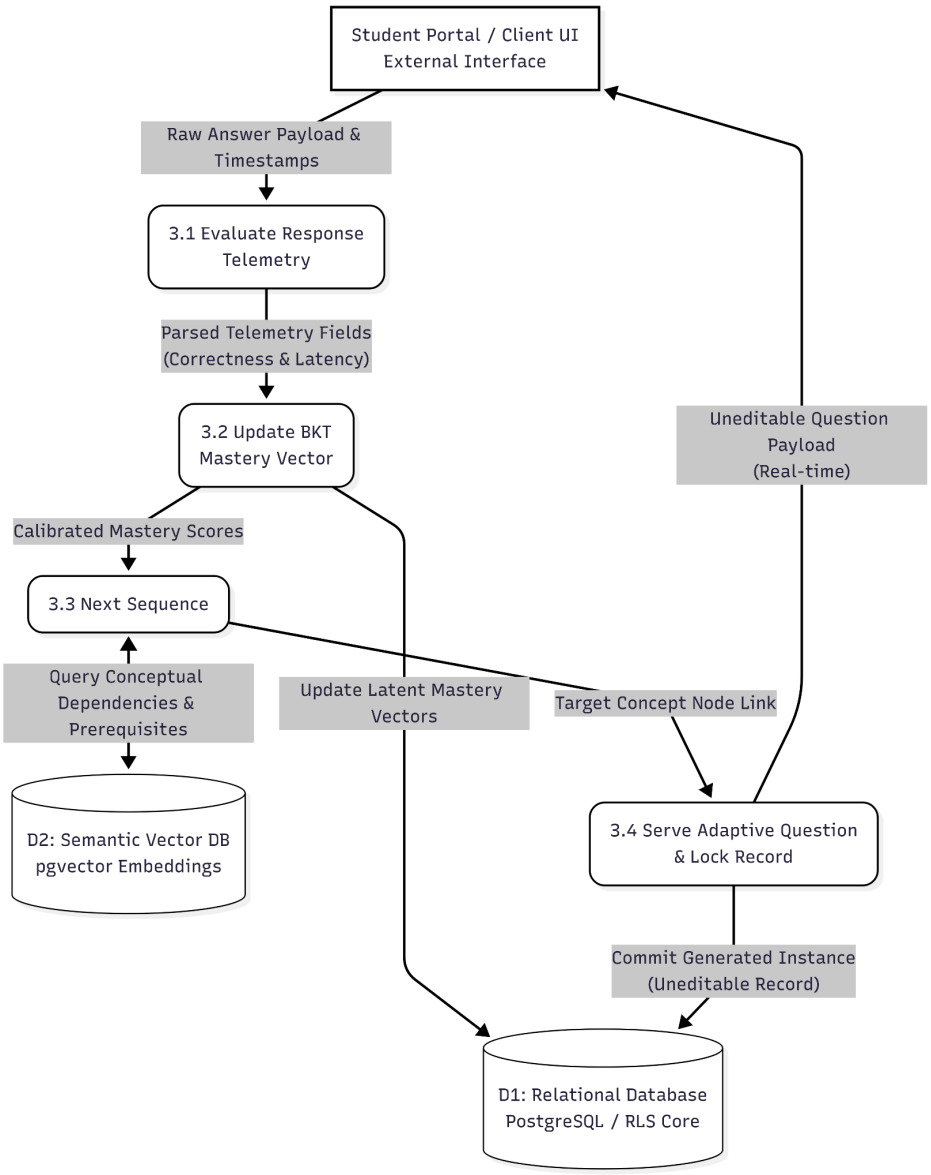


Figure 5. *Level 2 AI Process*

TOOL TO TEST FOR ACCURACY

This study implements a specialized evaluation framework to validate the structural integrity and semantic precision of LearnSync AI's diagnostic modules. Because the platform relies on dynamic predictive microservices and cognitive modeling rather than static code paths, traditional functional testing alone is insufficient. The testing architecture therefore isolates the machine learning logic to verify concept tracking, diagnostic consistency, and predictive routing prior to live deployment. To measure predictive accuracy, the microservices undergo continuous model validation. The Mastery Placement Engine Evaluation uses a confusion matrix to assess whether a student will master or fail an upcoming concept based on historical performance. The matrix categorizes outputs as true positives (correct mastery identification), true negatives (correctly identifying not yet mastered), false positives (incorrect mastery labeling), and false negatives (incorrectly flagging struggle). From these, the system calculates precision (minimizing false positives), recall (ensuring struggling students are not overlooked), and an F1-score for overall diagnostic balance. Concurrently, the Cognitive Tracking Accuracy Evaluation employs mean absolute error to test Bayesian Knowledge Tracing (BKT) accuracy, calculating the average statistical distance between predicted correctness probability and actual student performance across practice sessions. This ensures the cognitive engine accurately mirrors learning progression and prevents algorithmic drift.

Complementing algorithmic validation, black-box testing verifies the input-output accuracy of backend microservices and API pipelines without

inspecting source code. Semantic Parser Testing confirms that unstructured instructor uploads yield a JSON quiz payload matching the document's schema. pgvector Query Validation ensures the PostgreSQL similarity operator returns appropriate semantic matches within a target threshold for Free Recall evaluations. Routing Logic Validation verifies that a string of incorrect answers correctly triggers BKT to route the student to a lower-tier concept or remediation track. CollabShare Permission Validation confirms that Supabase RLS policies keep shared spaces strictly read-only. Finally, RBAC Testing validates multi-tenant data accuracy, ensuring instructors access aggregated dashboards while student tokens remain locked out of administrative boundaries.

The testing lifecycle concludes with the ISO/IEC 25010 Quality Model Survey, a descriptive questionnaire administered to 100 institutional respondents to evaluate production readiness from an end-user perspective. This tool measures four core quality dimensions. Functional Suitability assesses the real-world accuracy of the generative AI parser, CollabShare RLS locks, and adaptive practice tests. Usability focuses on minimizing cognitive load and ensuring PlayMe elements (XP, study streaks, badges) remain intuitive. Performance Efficiency tracks backend response times and Supabase database latency under concurrent high-traffic exam submissions. Security evaluates strict Row-Level Isolation, protection of student profiles across virtual classrooms, and prevention of unauthorized write permissions during peer workspace sharing.

POPULATION AND SAMPLING

The population for this study consists of students and faculty members within Pamantasan ng Lungsod ng Pasig (PLP) who regularly engage with traditional course materials and classroom monitoring frameworks. The study particularly focuses on academic programs that currently rely on static document distribution, unorganized study metrics, and manual, uncoordinated study groups, as these environments are most likely to benefit from digital interventions such as AI-powered parsing tools and automated adaptive pathways (Li et al., 2025; Yu et al., 2022). To isolate an ideal test group, a purposive sampling strategy is deployed to identify individuals directly situated within technical learning tracks who demonstrate an active readiness to adopt novel educational technology frameworks (Alruwais & Zakariah, 2023). Specifically, the recruitment criteria isolates instructors open to automating digital material parsing workflows, alongside undergraduate students enrolled in concept-heavy courses that mandate intense independent study paired with collaborative workspace sharing.

It is anticipated that selected classes within the College of Computer Studies, specifically micro-targeted toward Bachelor of Science in Information Technology (BSIT) and Bachelor of Science in Computer Science (BSCS) tracks, will serve as the pilot partner environment for testing the system. Within these classes, the active sample size is structured across two primary operational cohorts, beginning with a student cohort of approximately 100 undergraduate college students aged 18–25 who will act as primary test users. This cohort is responsible for executing independent review sessions and providing user

feedback concerning interface usability, adaptive BKT routing logic efficiency, and gamification layout clarity. Additionally, a faculty cohort comprising 3 to 5 active instructors will evaluate the system's administrative performance, reporting modules, and data isolation capabilities. This multi-perspective sampling strategy ensures that the gathered evaluation data accurately reflects both frontline student interactions and professional academic oversight, helping to identify potential operational bugs, system latency challenges, and precise areas for software performance tuning prior to a broader rollout across the entire institution (Alruwais & Zakariah, 2023; Yu et al., 2022).

DATA COLLECTION METHOD

The study will utilize both secondary and primary data to inform the design and evaluation of the AI-powered adaptive learning and automated tracking system (Alruwais & Zakariah, 2023; Li et al., 2025). Secondary data will be collected from research literature on cognitive student modeling, established adaptive learning frameworks, pedagogical question routing rules, and AI-driven educational applications (Alruwais & Zakariah, 2023; Li et al., 2025). These sources will directly inform the creation of automated parsing templates, structured concept dependency maps, and adaptive response parameters that are accurate, logical, and educationally relevant (Yu et al., 2022).

Primary data will be obtained through structured feedback loops with students, collection of actual course syllabi and instructional material files,

student feedback on traditional study spaces, and observation of group dynamics within the participating academic cohorts. The data collection process will simulate real-world academic operations by entering sample student information, course outlines, text-heavy documents, and test responses directly into the system. This operational approach allows the researchers to thoroughly test backend pipeline functionality, baseline algorithmic error margins, and frontend response times while identifying potential software bottlenecks without compromising actual student grades or classroom standings (Hafdi & El Kafhali, 2025.; Naqvi et al., 2025)

DATA ANALYSIS PROCEDURE

Data analysis will focus on evaluating the projected performance of the AI system, user engagement markers, and overall platform reliability (Alruwais & Zakariah, 2023; Hafdi & El Kafhali, 2025.). Preprocessing of text and telemetry data will include standardizing raw student response metrics, normalizing key concept terms extracted from uploaded files, and mapping uniform performance logs to ensure structural consistency and system calculation accuracy (Alruwais & Zakariah, 2023; Hafdi & El Kafhali, 2025.).

The AI components, including the generative quiz factory and the adaptive routing engine, are projected to undergo testing to evaluate their capacity for accurate content extraction, context-sensitive question delivery, and maintenance of logical pedagogical coherence (Li et al., 2025; Yu et al., 2022). The automated

tracking system will also be analyzed for its expected effectiveness in improving student compliance with practice schedules, including tracking optimal study frequencies, gamification progression patterns, and workspace cloning frequencies based on simulated user behavioral telemetry (Hafdi & El Kafhali, 2025.).

Statistical analysis will be conducted using metrics such as accuracy, precision, recall, and Mean Absolute Error (MAE) to assess the mathematical precision of the underlying machine learning models. Additionally, user acceptance and usability testing will be carried out, where participants will evaluate interface navigability, ease of access to personalized practice modules, reliability of gamification rewards, and the helpfulness of analytics dashboards. Insights extracted from this combined quantitative and qualitative evaluation framework will steer the iterative development cycles of the application, thereby guaranteeing that the deployed system successfully fulfills the daily operational requirements of both student learners and academic educators across the university ecosystem (Naqvi et al., 2025; Nugraha & Rachman, 2024).

ETHICAL CONSIDERATIONS

The deployment of LearnSync AI follows a rigorous ethical framework to safeguard participant welfare, protect data, and promote equitable education. Because the platform relies on real-time data collection and predictive models, all sensitive telemetry is secured within Supabase PostgreSQL multi-tenant tables,

with raw individual data encrypted at rest. Visibility is strictly governed by the RBAC decryption engine. Instructors see only aggregated analytics for their classes, while individual student profiles remain isolated. This ensures the system identifies academic learning gaps without enabling invasive surveillance, thereby preserving digital dignity.

In alignment with student autonomy, CollabShare workspace sharing is entirely voluntary. The sharing student initiates the sync token, and the system applies strict, immutable PostgreSQL Row-Level Security (RLS) policies on all shared flashcards, recall logs, and tests. Peer students can clone study materials to run them under their own profiles, but are mathematically prohibited from modifying the original owner's workspace, preventing plagiarism and bad-faith data alteration.

The study also mandates a transparent, voluntary consent process before system interaction. Participants receive explicit documentation on what data is captured (e.g., time-on-task latency, quiz accuracy, free recall inputs) and how it drives BKT routing. They sign a form acknowledging that immutable telemetry is used solely for educational personalization within this application, and they may withdraw at any time without academic penalty.

A core operational priority is preventing systemic bias. The research acknowledges external limitations such as varying device capabilities or unstable internet connectivity. To mitigate this, the platform uses an asynchronous Supabase real-time pipeline, ensuring students with slower connections are not

penalized by timeout errors or latency metrics during adaptive assessments, maintaining a fair, balanced, and highly equitable learning environment for every user.

SUMMARY

The quantitative research design combines descriptive-evaluative and quasi-experimental methodologies, making it uniquely suited for verifying both the algorithmic precision and the educational impact of LearnSync AI. By converting raw system interactions into numeric data, the design eliminates human bias from the evaluation process. The quasi-experimental component captures changes in student concept mastery through pre-test and post-test tracking within the independent practice environment. Concurrently, the descriptive-evaluative framework provides structural controls to monitor and correlate backend telemetry (including question correctness ratios, response latencies, and workspace cloning logs) proving that the adaptive models respond accurately to live student behaviors.

To execute these complex requirements, the Agile Scrum framework offers operational flexibility that traditional linear models lack. Sequential frameworks cannot effectively manage the unpredictable design, testing, and training pathways of machine learning microservices and NLP pipelines. Scrum's iterative sprints allow the development team to build, evaluate, and tune backend data paths in short, manageable cycles. This adaptive configuration directly supports

the three-tier cloud architecture, ensuring that role-based access tokens, row-level database partitioning (RLS), and asynchronous caching layers integrate seamlessly to handle high-velocity data streams without system drops or penalizing students operating on slower internet connections.

The testing lifecycle combines automated validation with real-world user surveys to demonstrate absolute system readiness for institutional deployment. Black-box testing independently verifies microservice input-output behaviors and multi-tenant role restrictions without requiring manual source-code audits. This technical validation phase is followed by a 2-to-4-week pilot study with 100 undergraduate IT/CS students and 3–5 faculty instructors at Pamantasan ng Lungsod ng Pasig (PLP). Their live interactions are evaluated through an ISO/IEC 25010 Quality Model survey, successfully quantifying performance across functional suitability, usability, performance efficiency, and data security. Ultimately, this unified methodology provides a reliable, replicable engineering blueprint for safely transforming unstructured course inputs into fair, personalized, and fully auditable educational experiences.

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